

Bridging Disciplines for a New Era in Physical AI

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ABSTRACT

Physical AI is extending models from the digital world into performing tasks in the real world. Robots and autonomous cars are common examples of physical AI agents. In this workshop, we aim to go beyond containing physical AI in those agents and ask *what if all objects and materials we interact with were intelligent?* We aim to form an understanding of the opportunities and challenges of *extending agents* to include objects and materials that can adapt to users needs, i.e., change shape, firmness, color, tactile properties, etc. This broad vision, which is challenging to achieve, is related to many active research areas, e.g., programmable matter, modular robotics, soft robotics, smart materials, shape-changing interfaces, or radical atoms, and has homes in many disciplines, incl. mechanical engineering, robotics, material science, computer science. Many new approaches are being developed in the individual disciplines that together might be the start of a new era for what we like to call *extended physical AI*. In this workshop, we bring perspectives from these different disciplines together to exchange new approaches to longstanding challenges (e.g., actuation, computational design, fabrication, control), exchange tacit knowledge, discuss visions for future applications, map the new grand challenges, and inspire the next generation of physical AI research.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Physical AI, Programmable Matter, Metamaterials, Soft Robotics

ACM Reference Format:

Alexandra Ion, Carmel Majidi, Lining Yao, and Amir H. Alavi. 2024. Bridging Disciplines for a New Era in Physical AI. In *The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24)*, October 13–16, 2024, Pittsburgh, PA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3672539.3686704>

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UIST Adjunct '24, October 13–16, 2024, Pittsburgh, PA, USA

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ACM ISBN 979-8-4007-0718-6/24/10

<https://doi.org/10.1145/3672539.3686704>

1 INTRODUCTION

The world around us is inherently physical, yet adaptive interfaces focus mostly on digital content and representations. Physical AI is moving in the right direction by creating models that can understand instructions and to perform physical tasks in the real world [10]. Imagine a future where the physical objects surrounding us dynamically adapt to our needs, by not only adjusting their user interfaces to fit the context but also by transforming their physical features, material properties, and affordances—instantly becoming exactly what is required in that moment.

Creating such intelligent agents requires adaptive physical architectures. There are already active research areas developing such physical interfaces known as, e.g., programmable matter, radical atoms [4], modular [9] or soft robotics [8], metamaterials [2, 3, 5], morphing matter [6] or, most common in HCI, shape-changing interfaces [1, 7]. They have in common that objects and materials dynamically adapt to user needs, tasks, or environments by changing their properties, which is a notoriously difficult task as it needs strong actuation in ideally tiny form factors. This requires careful design of the objects and materials themselves to embed sensors to perceive users and their environments, to transform that input into desired output by self-adapting its properties. Achieving this vision involves contributions from many fields as each discipline brings unique insights into different challenges, e.g., user modeling, simulation, sensing, actuation, fabrication.

This workshop aims to unite experts from mechanical engineering, material science, computer science, HCI, and other related fields to share innovative approaches, discuss current challenges, envision future applications, and inspire new research directions in physical AI. UIST provides an ideal platform for this convergence, emphasizing the human interaction aspect of physical AI.

1.1 Scope

This workshop aims to bring together the UIST community with other disciplines that work on related topics. The workshop will cover topics such as actuation mechanisms, design and fabrication techniques, material properties and capabilities, human interaction with physical AI, and future applications and grand challenges in the field. The workshop is designed for researchers and practitioners from diverse backgrounds, including but not limited to mechanical engineering, robotics, material science, computer science, and human-computer interaction. Organizers will leverage their networks to attract a wide range of participants, fostering

connections between the UIST community and other researchers working on related topics.

2 PROPOSED FORMAT

2.1 Introductory Session

The workshop will kick off with an engaging introductory session. Organizers will deliver lightning talks, each lasting about 10 minutes, to share their perspectives on physical AI. Following this, workshop attendees will briefly present their work and perspectives for 2 minutes each, serving as introductions. Attendees are encouraged to bring samples of their work for a small demo session, allowing hands-on exploration of various projects.

2.2 Group Sessions

Participants will engage in four simultaneous group sessions, each lasting about 90 min with breaks between group sessions. Each group will be provided with materials to create one index card documenting each discussion point. We will try to secure assistants (e.g., students from CMU) to sketch the discussion point quickly during discussions for a visual representation. Participants will keep their cards for later clustering.

Group Session: Wild Applications. This session focuses on exploring and envisioning potential applications of physical AI. Participants will brainstorm what physical AI should be, how it would be helpful, and how people would want to use it, aiming to identify potential “killer apps.”

Group Session: New Opportunities. In this session, participants will discover and discuss new opportunities from other disciplines. They will share exciting discoveries and developments from their fields, such as new actuators, fabrication technologies, materials, etc. The goal is to foster cross-pollination of ideas, build on others’ insights, and explore how these new opportunities can be leveraged in different fields.

Group Session: Persistent Challenges. Participants will identify and map out persistent challenges across disciplines. They will discuss ongoing challenges in their fields and explore whether these challenges are also present in other fields or if solutions exist elsewhere. This exchange aims to uncover potential solutions and foster collaborative problem-solving.

Group Session: What If? This session will consider potential risks and sustainability issues. Participants will brainstorm potential negative outcomes (e.g., “grey goo” scenarios) and discuss how to mitigate these risks. They will also explore how to make physical AI sustainable and ethical in the long term.

2.3 Synthesis Session

In the final session, all participants will come together to present and synthesize discussion items and results. Each participant will briefly describe one item (e.g., a new fabrication technique from their field) and lay down their card. Other participants with similar or related cards will place them next to each other. If no related cards are available, a new cluster can be started. This will visualize common themes across research areas and disciplines, revealing unique perspectives, and mapping challenges and opportunities.

2.4 The End?

The workshop aims to spark new themes across disciplines and foster collaborations. Post-workshop, the goal is to write a position paper, possibly for submission to Nature Perspectives, summarizing the workshop’s findings and future directions. We will invite any interested participants to contribute to the paper.

3 FACILITATING AUDIENCE INTERACTION

To ensure active participation and meaningful interactions, the size of the groups will be limited to a maximum of 10 people. If more than 40 people sign up, additional sub-groups will be created to maintain small group dynamics. Hands-on demos and interactive discussions will be encouraged during the introductory session. Rotating group sessions will ensure participants engage with different perspectives and ideas. The synthesis session will consolidate insights and foster a collective vision for future research.

4 ORGANIZERS

Alexandra Ion is Sara Kiesler Assistant Professor at the Human-Computer Interaction Institute at Carnegie Mellon University, where she leads the Interactive Structures Lab. She focuses on on material discovery by broadening participation. To achieve this, Dr. Ion develops optimization-based interactive design tools that enable novices to contribute their creativity and experts to apply their intuition in order to foster the advancement of high-tech materials. Her research and expertise lies at the intersection of human-computer interaction, digital fabrication, metamaterials and computational design. Her website is <https://interactive-structures.org/>.

Carmel Majidi is the Clarence H. Adamson Professor in the Department of Mechanical Engineering at Carnegie Mellon University, where he leads the Soft Machines Lab. His career mission is to discover materials, hardware architectures, and fabrication methods that allow robots and machines to behave like soft biological organisms and be safe for contact with humans. The research aim of Dr. Majidi’s lab is to replace the bulky and rigid hardware in existing robots with soft, lightweight, and deformable technologies that match the functionality of natural biological tissue. His website is <http://sml.me.cmu.edu/>

Lining Yao is an Assistant Professor in the Department of Mechanical Engineering at the University of California, Berkeley, where she directs the Morphing Matter Lab. Dr. Yao’s research explores the positive impact of active and morphing materials on sustainable design across different scales and contexts. Her work focuses on discovering and studying morphing material mechanisms, as well as algorithms for computational design and fabrication pipelines. Her website is <https://morphingmatter.org/>.

Amir H. Alavi is an Assistant Professor in the Department of Civil and Environmental Engineering at the University of Pittsburgh, where he leads the Intelligent Structures and Architected Materials Research and Testing (ISMART) Lab. Dr. Alavi is passionate about conducting fundamental research to advance the science of “systems engineering” and develop integrative approaches for the exploration, design, fabrication, and computational modeling of novel engineered material and structural systems across multiple scales. His website is <https://sites.pitt.edu/~alavi/>.

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